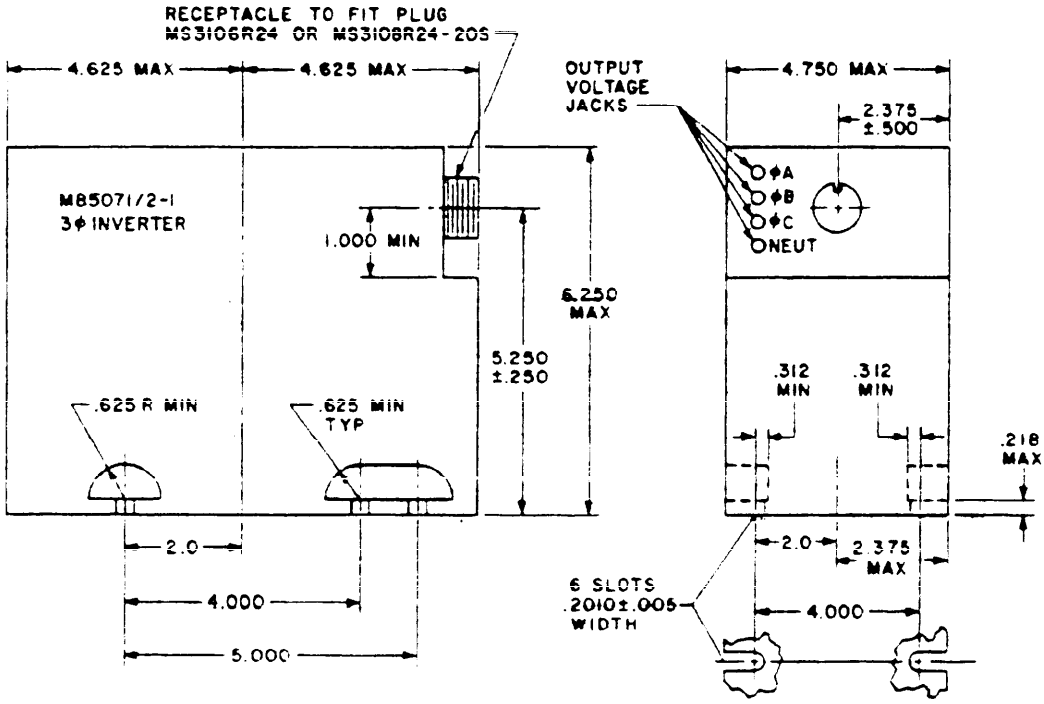
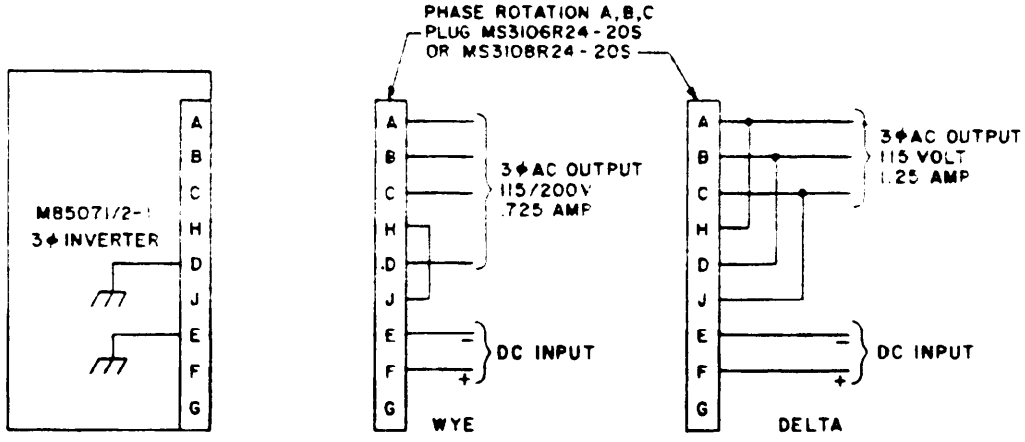
MILITARY SPECIFICATION SHEET

INVERTER, AIRCRAFT, DC TO AC,
STATIC, 250VA, 3-PHASE AND
250 VA 1-PHASE, MULTIPURPOSE, CLASS B

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense

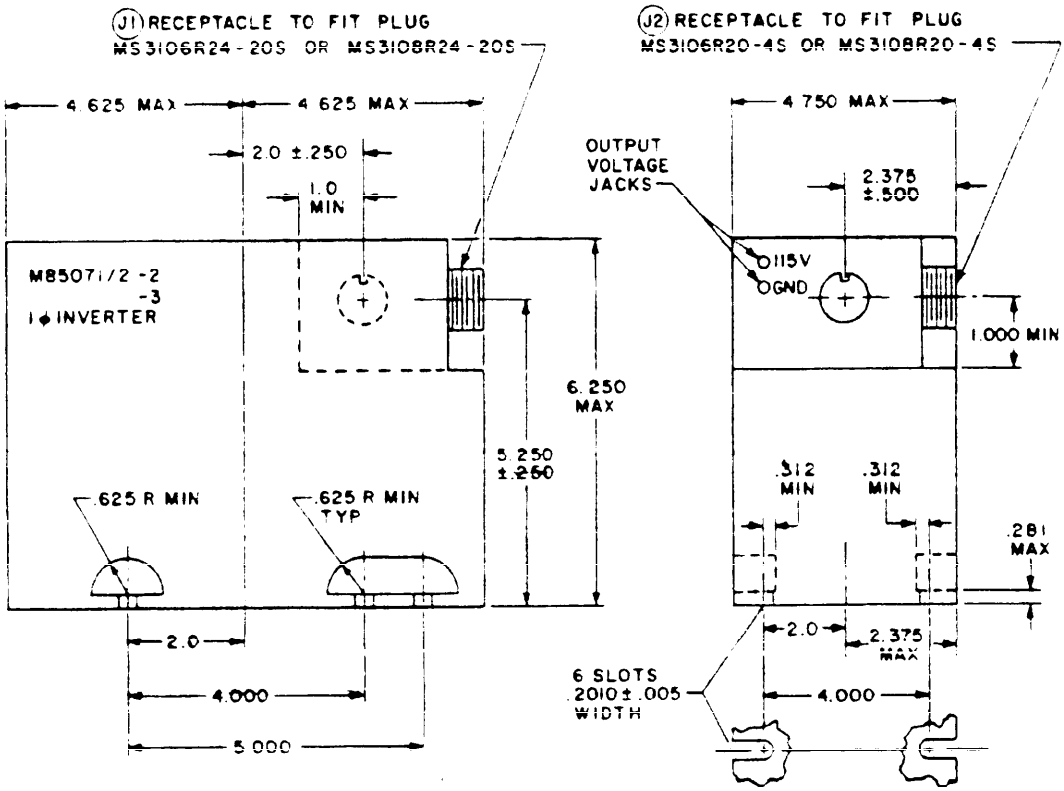
The complete requirements for procuring the inverters described herein shall consist of this document and the latest issue of Specification MIL-I-85071(AS).

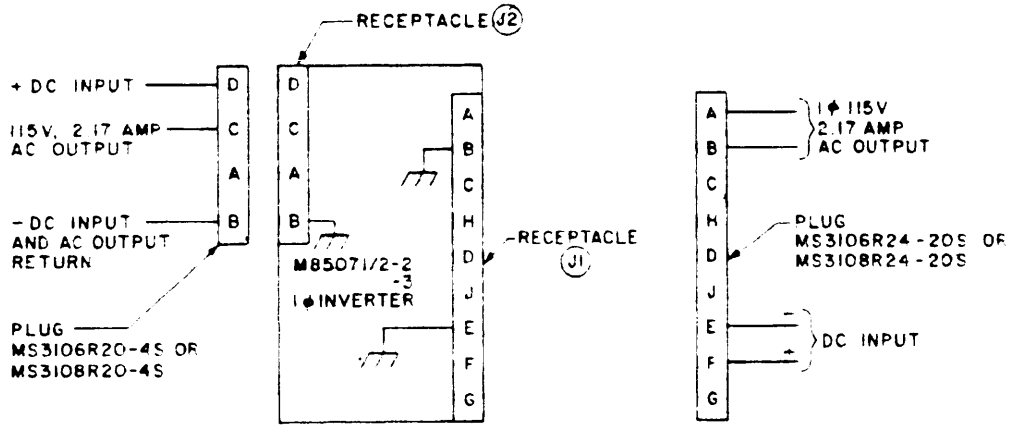




CONNECTIONS EXTERNAL TO THE INVERTER NECESSARY FOR 3Φ OPERATION

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CONNECTIONS EXTERNAL TO THE INVERTER NECESSARY FOR 1φ OPERATION

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DESCRIPTION

PART NO.	POWER RATING	POWER FACTOR	MAX WEIGHT POUNDS	SUPERSEDING
M85071/2-1	250 VA, 3Ø	.75 MIN LAG	13	MS21983-3
M85071/2-2	250 VA, 1Ø	.95 MIN LEAD	12	MS21983-4
M85071/2-3	250 VA, 1Ø LOW INPUT VOLTAGE		12	

REQUIREMENTS:

1. The inverter shall operate under the following conditions:

For any combination of conditions listed within each section below, the output voltage and frequency shall remain within the indicated limits	OUTPUT LIMITS LIMITS APPLY TO EACH PHASE		
	RMS VOLTAGE	FREQ (Hz)	NOTES
A. NORMAL INPUT VOLTAGE			
1. Input voltage varied between 26 and 29 volts 2. Load varied from zero to full load 3. Ambient temperature-altitude per Class B 4. Load power factor varied from .75 lag to .95 lead 5. Unit mounted in any position	Low 112.5 High 117.5	Low 393 High 407	
B. ABNORMAL INPUT VOLTAGE			
6. All variations above except input voltage between 19.5 and 26 volts and load from 0 to full rated load current.	Low 108 High 117.5	Low 393 High 407	
C. OVERLOAD			
7. Variations 1 through 5 above except load varied from rated full load current to 200% rated full load current for 5 seconds	Low 50 High 117.5	Low 393 High 407	
D. LOW INPUT VOLTAGE			
8. Input voltage varied between 9 and 19.5 volts for three minutes 9. Load impedance varied from infinity to 53 ohms for the -2 inverter and from infinity to 159 ohms/phase for the -1 inverter 10. Ambient temperature varied between -35°C and +40°C at sea level with inverter fully warmed up and between -35°C and +85°C at sea level with inverter not previously warmed up 11. Variations 4 and 5 above	Low 45 High 117.5	Low 393 High 407	Not applicable to -3 inverter
12. Variations 8, 10 and 11 above 13. Load impedance varied from infinity to 53 ohms	Low 108 High 117.5	Low 393 High 407	Not applicable to -1 and -2 inverters

2. The -1 inverter shall operate either in the wye or delta mode when external connections are made as shown in the connecting diagram.

3. Inverter samples 3 and 4 shall be tested for conformance to requirements in Section D above for qualification.

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THE FOLLOWING REQUIREMENTS APPLY TO THE -3 INVERTER ONLY:

4. The -3 inverter shall meet the requirements of the -2 inverter in addition to the -3 low input voltage operating requirements.

5. The total harmonic and extraneous content shall not exceed 8% of the fundamental under any rated operating condition with input voltage between 9 and 19.5 volts.

6. The inverter efficiency shall be 50% minimum at input voltages between 9 and 19.5 volts, sea level altitude, $20^{o} \pm 10^{o}C$ ambient temperature, 53 ohm load and unity power factor.

7. The inverter shall be subjected to 1250 consecutive cycles of operation consisting of one minute at 9 volts input and 53 ohm load and 4 minutes at 28 volt input and 250VA load. This test shall be conducted at room temperature with unity power factor loads. At the conclusion of the test, the inverter shall be tested for compliance with requirements for voltage and frequency regulation at room ambient, paragraph 4.7.2.2.1. For qualification, this test shall be conducted on the number 4 sample.

NOTES:

1. All dimensions are in inches.

2. Unless otherwise specified, tolerances: $\pm$ .015 inch.

Preparing Activity
Navy AS
Project No. 6130-N202-2

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-I-85071/2 (AS)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

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DATE

DD FORM 1 OCT 76 1426 Replaces edition of 1 Jan 72 which may be used. S/N 0102-LF-001-4260

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